

EE22005 Engineering Practice and Design

NEW BOARD — Thermal Calculations

4-Layer Design | With Thermal Design Considerations
Synchronous Buck Converter VIN=+12V VOUT=5V ID=5A f=300kHz
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1. Board & Component Specifications

Parameter	Value	Source
Board layers	4 (F.Cu / GND / VIN(+12V) / B.Cu)	KiCad Board Setup — SC7.5
F.Cu copper thickness	0.07mm (2oz)	SC7.5 Board Setup
GND / VIN inner layers	0.035mm (1oz) each	SC7.5 Board Setup
B.Cu copper thickness	0.7mm	SC7.5 Board Setup
PCB total thickness	2.3mm	SC7.5 Board Setup
Board area	50 × 45mm = 22.5cm ²	KiCad Edge.Cuts
All trace widths	0.5mm	KiCad — improved from 0.2mm
Thermal vias per MOSFET	12 (3 per corner of mounting tab pad)	KiCad
Via drill diameter	0.3mm	PCBWay minimum [3]
Via pad diameter	0.6mm	Annular ring 0.15mm = PCBWay min [3]
Via plating thickness	25µm standard	PCB fabrication standard
Heatsink	Fischer Elektronik SK104 38.1mm	Specified
Thermal paste	Wakefield-Vette 120 compound	Applied between case and heatsink
θCS with paste	0.50°C/W	Direct from IRFZ44N datasheet [1]
θSA (Fischer SK104 38.1mm)	3.2°C/W	Fischer specification

2. IRFZ44N Datasheet Values Used [1]

Parameter	Symbol	Value	Units
Junction-to-case thermal resistance	θ_{JC}	1.5 max	°C/W
Case-to-sink, flat GREASED surface	θ_{CS}	0.50	°C/W — directly from datasheet table
Max junction temperature	$T_J \text{ max}$	175	°C
Drain-source on-resistance	$R_{DS(on)}$	17.5	m Ω
Rise time	t_r	60	ns
Fall time	t_f	60	ns
Continuous drain current	I_D	49	A

3. Power Dissipation — Q1 (same as old board)

Power dissipation is determined by the circuit, not the PCB layout — same values for both boards.

$$P_{\text{cond}} = 5^2 \times 0.0175 \times 0.5 = 0.22 \text{ W} \quad [1]$$

$$P_{\text{sw}} = 0.5 \times 12 \times 5 \times 120 \times 10^{-9} \times 300,000 = 1.08 \text{ W} \quad [1]$$

$$PD_{Q1} = 0.22 + 1.08 = 1.30 \text{ W}$$

4. Thermal Resistance Calculations — New Board

4.1 Thermal Via Resistance (Rule 2) [4]

Formula: $\theta_{VIA} = L / (\lambda_{Cu} \times \pi \times [r^2 - (r - t_{\text{plating}})^2])$

Via length = FULL board thickness (2.3mm) — standard through-hole vias in KiCad.

$\lambda_{Cu} = 401 \text{ W/m}\cdot\text{K}$ [4]

$$\begin{aligned}
 r &= 0.15 \times 10^{-3} \text{ m} && (0.3\text{mm drill} \div 2) \\
 t_{\text{plating}} &= 25 \times 10^{-6} \text{ m} && (25\mu\text{m standard copper plating}) \\
 L &= 2.3 \times 10^{-3} \text{ m} && (\text{full board thickness}) \\
 \text{Numerator: } &2.3 \times 10^{-3} \\
 \text{Denominator: } &401 \times \pi \times [(0.15 \times 10^{-3})^2 - (0.125 \times 10^{-3})^2] \\
 &= 401 \times \pi \times [2.25 \times 10^{-8} - 1.5625 \times 10^{-8}] \\
 &= 401 \times \pi \times 6.875 \times 10^{-9} \\
 &= 401 \times 2.160 \times 10^{-8} \\
 &= 8.661 \times 10^{-6} \\
 \theta_{\text{VIA_single}} &= 2.3 \times 10^{-3} / 8.661 \times 10^{-6} = 265.6 \text{ }^{\circ}\text{C/W per via} \\
 \theta_{\text{VIAS (12 parallel vias)}} &= 265.6 / 12 = 22.1 \text{ }^{\circ}\text{C/W}
 \end{aligned}$$

4.2 Board Surface to Ambient [4]

$$\begin{aligned}
 \text{Board area} &= 50\text{mm} \times 45\text{mm} = 22.5 \text{ cm}^2 \\
 \theta_{\text{SA_plane}} &= 1000 / 22.5 = 44.4 \text{ }^{\circ}\text{C/W}
 \end{aligned}$$

4.3 Path 1 — Heatsink with Thermal Paste (Primary Path)

$\theta_{\text{CS}} = 0.50^{\circ}\text{C/W}$ taken DIRECTLY from IRFZ44N datasheet [1]: 'R θ_{CS} Case-to-Sink, Flat, Greased Surface = 0.50°C/W '

$$\begin{aligned}
 \theta_{\text{JA_path1}} &= \theta_{\text{JC}} + \theta_{\text{CS}} + \theta_{\text{SA_heatsink}} \\
 &= 1.5 + 0.50 + 3.2 \\
 \theta_{\text{JA_path1}} &= 5.2 \text{ }^{\circ}\text{C/W}
 \end{aligned}$$

4.4 Path 2 — Thermal Via Array → Inner GND Plane (Secondary Path)

$$\begin{aligned}
 \theta_{\text{JA_path2}} &= \theta_{\text{JC}} + \theta_{\text{VIAS}} + \theta_{\text{SA_plane}} \\
 &= 1.5 + 22.1 + 44.4 \\
 \theta_{\text{JA_path2}} &= 68.0 \text{ }^{\circ}\text{C/W}
 \end{aligned}$$

4.5 Combined θ_{JA} — Parallel Heat Paths

$$\begin{aligned}
 1/\theta_{\text{JA}} &= 1/\theta_{\text{JA_path1}} + 1/\theta_{\text{JA_path2}} \\
 &= 1/5.2 + 1/68.0 \\
 &= 0.1923 + 0.0147 \\
 &= 0.2070 \\
 \theta_{\text{JA_new}} &= 1/0.2070 = 4.83 \text{ }^{\circ}\text{C/W}
 \end{aligned}$$

4.6 Junction Temperature

$$\begin{aligned}
 T_{\text{J}} &= T_{\text{A}} + (\text{PD} \times \theta_{\text{JA}}) \\
 &= 25 + (1.30 \times 4.83)
 \end{aligned}$$

$T_{\text{J_new}} = 31.3^{\circ}\text{C}$ Margin from $T_{\text{J max}} 175^{\circ}\text{C}$ [1]: 143.7°C

5. Old vs New Board — Full Thermal Comparison

Parameter	Old Board	New Board	Source	Improvement
Trace width	0.2mm	0.5mm	KiCad	2.5× wider
Copper thickness	35µm (1oz)	70µm (2oz)	Board Setup	2× thicker
Thermal vias	None	12 (3/corner)	KiCad	New path
θJC	1.5°C/W	1.5°C/W	[1] Fixed	—
θCS	1.8°C/W	0.50°C/W	[1] greased	3.6× better
θSA heatsink	6.5°C/W	3.2°C/W	Fischer spec	2.0× better
θFR4 / θVIAS	53.3°C/W	22.1°C/W	$\lambda=0.3/401 \text{ W/m}\cdot\text{K}$	2.4× better
θSA board/plane	30.6°C/W	44.4°C/W	1000/area [4]	Smaller board
θJA_path1	9.8°C/W	5.2°C/W	Calculated	1.9× better
θJA_path2	85.4°C/W	68.0°C/W	Calculated	1.3× better
θJA COMBINED	8.79°C/W	4.83°C/W	Parallel paths	1.8× better
TJ (PD=1.30W, TA=25°C)	36.4°C	31.3°C	Calculated	5.1°C cooler
TJ margin from 175°C	138.6°C	143.7°C	[1]	+5.1°C

References

[1] Infineon Technologies, "IRFZ44NPbF HEXFET Power MOSFET," Datasheet Rev. PD-94787B. Available: https://www.infineon.com/dgdl/Infineon-IRFZ44N-DataSheet-v01_01-EN.pdf

[2] Texas Instruments, "UCC27211 120V 3.7A/4.5A Half-Bridge Driver," Datasheet Rev. H. Available: <https://www.ti.com/lit/ds/symlink/ucc27211.pdf>

[3] PCBWay, "PCB Manufacturing Capabilities." Available: <https://www.pcbway.com/capabilities.html>

[4] A. Mohammadi, "PCB Design and Manufacture: Thermal Design," EE22005, University of Bath, 2025.

[5] A. Mohammadi, "PCB Design and Manufacture: Design for Manufacturing," EE22005, University of Bath, 2025.

[6] A. Mohammadi, "PCB Design and Manufacture: Signal Integrity," EE22005, University of Bath, 2025.